

Q1: Integrated / Capstone Design [R]

Company	Sector	Revenue	ProfitMgn	Employees	MarketCap
C1	IT	420	8.2	1500	8500
C2	Banking	580	15.6	3200	12000
C3	Healthcare	720	11.4	2100	9800
C4	Retail	850	9.3	4200	7600
C5	Energy	980	18.5	2800	14500
C6	IT	1150	20.4	5100	18200
C7	Banking	1320	17.8	6800	21500
C8	Healthcare	1480	14.2	3900	16800
C9	Retail	1650	10.5	7200	13200
C10	Energy	1820	22.1	4500	24500
C11	IT	2050	24.5	8100	28500
C12	Banking	2280	19.8	9500	32500
C13	Healthcare	2550	16.7	6200	24800
C14	Retail	2840	13.4	10300	19200
C15	Energy	3150	26.2	7200	38200
C16	IT	3480	28.1	11800	42500
C17	Banking	3860	21.5	13200	46800
C18	Healthcare	4250	18.3	8600	35600
C19	Retail	4680	15.2	14500	29800
C20	Energy	5200	30.4	16000	52000

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.

Q3: Cartesian vs. Polar Coordinates [R]

Month	Coal	Gas	Solar	Wind	Hydro
Jan	620	310	45	80	140

Feb	580	295	60	95	150
Mar	590	305	90	120	160
Apr	605	320	130	150	170
May	640	340	180	170	180
Jun	670	355	230	185	195
Jul	690	360	250	195	205
Aug	720	370	245	190	210
Sep	700	365	210	175	200
Oct	680	350	160	150	185
Nov	650	335	90	110	165
Dec	630	320	50	90	150

The following dataset contains monthly electricity generation from different energy sources. Develop visualizations using Cartesian Coordinates and Polar Coordinates. Compare both visualizations and determine which coordinate system provides better analytical insight. Explain your design decisions.

Q2: Color Scales for Multi-Variable Data [R]

City	Temp	Humidity	Rainfall	AQI
Delhi	44	22	6	365
Mumbai	35	84	280	110
Chennai	38	78	220	125
Kolkata	40	72	190	180
Hyderabad	39	48	85	160
Bangalore	31	66	120	72
Pune	33	58	95	84
Ahmedabad	43	26	2	290
Jaipur	45	18	1	320
Lucknow	39	42	35	240
Bhopal	38	36	48	175
Patna	40	45	72	220
Goa	32	89	360	58

Kochi	30	91	410	52
Nagpur	41	30	12	270
Indore	37	34	20	190
Surat	36	76	180	145
Visakhapatnam	34	82	240	98
Coimbatore	29	68	130	60
Mysuru	28	64	145	55

T17	810000	490	46	7.6
T18	1050000	560	50	8
T19	1350000	640	55	8.5
T20	1800000	720	60	9

A software company has recorded project statistics for twenty software development teams. Design a visualization that requires the use of a nonlinear axis transformation to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale.

An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents Temperature, Air Quality Index (AQI), Rainfall, and Humidity using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

Q1. Integrated / Capstone Design [R]

Q4: Nonlinear Axis for Skewed Data [R]

Team	LinesOfCode	Bugs	Devs	Complexity
T1	12000	35	5	2.1
T2	18000	52	7	2.4
T3	25000	60	9	2.7
T4	35000	72	10	3
T5	48000	88	12	3.2
T6	62000	110	14	3.5
T7	81000	125	16	3.8
T8	96000	142	18	4.1
T9	120000	165	20	4.5
T10	150000	188	22	4.8
T11	185000	205	25	5.1
T12	240000	240	28	5.4
T13	310000	280	30	5.8
T14	390000	320	34	6.2
T15	480000	365	38	6.6
T16	620000	420	42	7.1

1. Objective

To develop an integrated visualization for **Revenue, Profit Margin, Market Capitalization, Employee Strength and Industry Sector** of 20 companies using suitable aesthetic mappings, coordinates, scales and colours.

2. R Code – Dataset

```
library(ggplot2)

company <- c("C1","C2","C3","C4","C5","C6","C7","C8","C9","C10",
            "C11","C12","C13","C14","C15","C16","C17","C18","C19","C20")

sector <- c("IT","Banking","Healthcare","Retail","Energy",
            "IT","Banking","Healthcare","Retail","Energy",
            "IT","Banking","Healthcare","Retail","Energy",
            "IT","Banking","Healthcare","Retail","Energy")

revenue <- c(420,580,720,850,980,1150,1320,1480,1650,1820,
            2050,2280,2550,2840,3150,3480,3860,4250,4680,5200)

profit <- c(8.2,15.6,11.4,9.3,18.5,20.4,17.8,14.2,10.5,22.1,
            24.5,19.8,16.7,13.4,26.2,28.1,21.5,18.3,15.2,30.4)

employees <- c(1500,3200,2100,4200,2800,5100,6800,3900,7200,4500,
            8100,9500,6200,10300,7200,11800,13200,8600,14500,16000)

marketcap <-
c(8500,12000,9800,7600,14500,18200,21500,16800,13200,24500,
            28500,32500,24800,19200,38200,42500,46800,35600,29800,52000)

data1 <- data.frame(company, sector, revenue, profit,
```

employees, marketcap)

- Qualitative colour scale represents Industry Sector.

3. Aesthetic Mappings

Variable	Aesthetic mapping
Revenue	X-axis
Profit Margin	Y-axis
Market Capitalization	Point size
Industry Sector	Colour
Company	Label
Employees	Label / additional information

4. Visualization

```
ggplot(data1,
  aes(x = revenue,
    y = profit,
    size = marketcap,
    colour = sector,
    label = company)) +
  geom_point(alpha = 0.75) +
  geom_text(vjust = -1, size = 3) +
  scale_size_continuous(name = "Market Capitalization") +
  scale_colour_brewer(palette = "Set2") +
  labs(title = "Integrated Company Performance Analysis",
    x = "Revenue",
    y = "Profit Margin (%)",
    colour = "Industry Sector") +
  theme_minimal()
```

5. Coordinate System

The **Cartesian coordinate system** is used because it provides direct comparison between Revenue and Profit Margin.

```
+ coord_cartesian()
```

6. Scale Selection

- Continuous scale is used for Revenue.
- Continuous scale is used for Profit Margin.
- Continuous point-size scale represents Market Capitalization.

7. Colour Palette

The **Set2 qualitative palette** is selected because Industry Sector is categorical.

```
scale_colour_brewer(palette = "Set2")
```

It provides clearly distinguishable colours and reduces confusion between sectors.

8. Observation

- Revenue generally increases with Market Capitalization.
- C20 has the highest Revenue and Market Capitalization.
- C20 also has the highest Profit Margin.
- IT, Banking, Healthcare, Retail and Energy companies can be distinguished using colour.
- Larger bubbles indicate companies with larger Market Capitalization.

9. Conclusion

The integrated scatter/bubble plot effectively represents multiple company-performance variables in one visualization. Cartesian coordinates, point size and a qualitative colour palette provide clear and easy comparison.

Q2. Color Scales for Multi-Variable Data [R]

1. Objective

To represent **Temperature, AQI, Rainfall and Humidity** for 20 Indian cities using suitable colour scales and explain the accessibility and interpretation of the selected palette.

2. R Code – Dataset

```
library(ggplot2)
```

```
city <- c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
  "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
  "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
  "Surat","Visakhapatnam","Coimbatore","Mysuru")
```

```
temp <- c(44,35,38,40,39,31,33,43,45,39,
          38,40,32,30,41,37,36,34,29,28)

humidity <- c(22,84,78,72,48,66,58,26,18,42,
             36,45,89,91,30,34,76,82,68,64)

rainfall <- c(6,280,220,190,85,120,95,2,1,35,
             48,72,360,410,12,20,180,240,130,145)

aqi <- c(365,110,125,180,160,72,84,290,320,240,
        175,220,58,52,270,190,145,98,60,55)

data2 <- data.frame(city,temp,humidity,rainfall,aqi)
```

3. Visualization Design

- A **scatter plot** is used:
- X-axis → Temperature
 - Y-axis → Rainfall
 - Colour → AQI
 - Point size → Humidity

```
ggplot(data2,
       aes(x = temp,
           y = rainfall,
           colour = aqi,
           size = humidity)) +
  geom_point(alpha = 0.8) +
  scale_colour_gradient(low = "yellow",
                       high = "red") +
  scale_size_continuous(range = c(3,10)) +
  labs(title = "Climate and Air Quality of Indian Cities",
       x = "Temperature (°C)",
       y = "Rainfall",
       colour = "AQI",
       size = "Humidity (%)") +
  theme_minimal()
```

4. Temperature Representation

Temperature is represented using the **X-axis**.

Higher values indicate hotter cities.

5. Rainfall Representation

Rainfall is represented using the **Y-axis**.

Cities with higher rainfall appear higher on the graph.

6. AQI Colour Scale

AQI is represented using a continuous colour gradient.

```
scale_colour_gradient(low = "yellow",
                     high = "red")
```

Lower AQI values appear toward yellow and higher AQI values toward red.

7. Humidity Representation

Humidity is represented using point size.

```
size = humidity
```

Larger points indicate higher humidity.

8. Colour Palette Justification

A sequential colour scale is suitable because AQI is numerical and has an ordered meaning.

The colour progression makes it easier to identify cities with relatively higher or lower AQI.

For accessibility, a colour scale with sufficient contrast should be used, and colour should not be the only method of communicating important information.

9. Observation

- Jaipur and Delhi have high temperatures and high AQI values.
- Kochi and Goa have high rainfall and humidity.
- Cities with larger points have higher humidity.
- Cities with higher AQI are shown with stronger colours.
- The visualization allows multiple environmental variables to be compared simultaneously.

10. Conclusion

The multi-variable visualization effectively represents Temperature, Rainfall, AQI and Humidity. Position, size and colour provide different channels for interpreting the environmental data.

Q3. Cartesian vs. Polar

Coordinates [R]

1. Objective

To visualize monthly electricity generation using **Cartesian and Polar coordinate systems**, compare both visualizations and identify the coordinate system that provides better analytical insight.

2. R Code – Dataset

```
library(ggplot2)
```

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
```

```
coal <- c(620,580,590,605,640,670,690,720,700,680,650,630)
```

```
gas <- c(310,295,305,320,340,355,360,370,365,350,335,320)
```

```
solar <- c(45,60,90,130,180,230,250,245,210,160,90,50)
```

```
wind <- c(80,95,120,150,170,185,195,190,175,150,110,90)
```

```
hydro <- c(140,150,160,170,180,195,205,210,200,185,165,150)
```

```
data3 <- data.frame(month,coal,gas,solar,wind,hydro)
```

3. Convert Data into Long Format

```
library(tidyr)
```

```
long_data <- pivot_longer(data3,  
                           cols = coal:hydro,  
                           names_to = "Source",  
                           values_to = "Generation")
```

```
long_data$month <- factor(long_data$month,  
                          levels = month)
```

4. Cartesian Coordinate Visualization

```
ggplot(long_data,  
       aes(x = month,  
           y = Generation,  
           colour = Source,  
           group = Source)) +
```

```
geom_line(linewidth = 1) +  
geom_point(size = 2) +  
labs(title = "Monthly Electricity Generation - Cartesian Coordinates",  
      x = "Month",  
      y = "Electricity Generation",  
      colour = "Energy Source") +  
theme_minimal()
```

5. Cartesian Design

- X-axis represents months.
- Y-axis represents electricity generation.
- Different colours represent energy sources.
- Lines show changes over time.

6. Cartesian Observation

- Coal has the highest generation throughout most months.
 - Solar generation increases from January toward July/August and then decreases.
 - Wind also increases during the middle of the year.
 - Gas remains relatively stable.
 - Cartesian coordinates make month-to-month comparisons very clear.
-

7. Polar Coordinate Visualization

```
ggplot(long_data,  
       aes(x = month,  
           y = Generation,  
           colour = Source,  
           group = Source)) +  
geom_line(linewidth = 1) +  
geom_point(size = 2) +  
coord_polar() +  
labs(title = "Monthly Electricity Generation - Polar Coordinates",  
      x = "Month",  
      y = "Electricity Generation",  
      colour = "Energy Source") +  
theme_minimal()
```

8. Polar Design

- Months are arranged around a circle.
- Generation is represented by radial distance.
- Different colours represent energy sources.

- The circular layout emphasizes the yearly cycle.

9. Polar Observation

- Seasonal patterns become visually noticeable.
- Solar generation shows a clear rise and fall during the year.
- Wind also shows seasonal variation.
- However, exact comparisons between individual months are harder.

10. Comparison

Feature	Cartesian	Polar
Month-to-month comparison	Excellent	Moderate
Seasonal pattern	Good	Excellent
Exact values	Easier	More difficult
Trend analysis	Excellent	Moderate
Cyclic pattern	Moderate	Excellent
Overall readability	High	Moderate

11. Better Coordinate System

Cartesian coordinates provide better analytical insight for this dataset.

12. Reason

The data needs comparison of electricity generation across months and energy sources. Cartesian coordinates make differences, trends and exact values easier to compare.

Polar coordinates are useful for showing the **cyclic nature of the year**, but they can make precise comparisons more difficult.

13. Conclusion

Cartesian coordinates are more suitable for detailed analytical comparison, while polar coordinates are useful for highlighting seasonal and cyclical patterns.

Q4. Nonlinear Axis for Skewed Data [R]

1. Objective

To design a visualization for highly skewed software project data using a **nonlinear axis transformation** and compare it with a linear scale.

2. R Code – Dataset

```
library(ggplot2)

team <- paste0("T",1:20)

lines <- c(12000,18000,25000,35000,48000,62000,81000,96000,
120000,150000,185000,240000,310000,390000,480000,
620000,810000,1050000,1350000,1800000)

bugs <- c(35,52,60,72,88,110,125,142,165,188,
205,240,280,320,365,420,490,560,640,720)

devs <- c(5,7,9,10,12,14,16,18,20,22,
25,28,30,34,38,42,46,50,55,60)

complexity <- c(2.1,2.4,2.7,3.0,3.2,3.5,3.8,4.1,
4.5,4.8,5.1,5.4,5.8,6.2,6.6,7.1,
7.6,8.0,8.5,9.0)

data4 <- data.frame(team,lines,bugs,devs,complexity)
```

3. Identify the Problem

Lines of Code range from:

12,000 to 1,800,000

This creates a highly skewed distribution.

A linear axis can cause smaller projects to appear compressed near the origin.

4. Linear Scale Visualization

```
ggplot(data4,
  aes(x = lines,
    y = bugs,
    size = devs,
    colour = complexity)) +
  geom_point(alpha = 0.8) +
  labs(title = "Software Teams - Linear Scale",
    x = "Lines of Code",
    y = "Number of Bugs",
    size = "Developers",
    colour = "Complexity") +
  theme_minimal()
```

5. Observation – Linear Scale

- Large projects occupy most of the horizontal range.
- Smaller projects are compressed toward the left.
- Differences between smaller teams are difficult to see.
- The distribution is strongly skewed.

6. Nonlinear Transformation

A **logarithmic transformation** is suitable because Lines of Code increase by a large multiplicative range.

```
ggplot(data4,
  aes(x = lines,
    y = bugs,
    size = devs,
    colour = complexity)) +
  geom_point(alpha = 0.8) +
  scale_x_log10() +
  labs(title = "Software Teams - Logarithmic Scale",
    x = "Lines of Code (Log Scale)",
    y = "Number of Bugs",
    size = "Developers",
    colour = "Complexity") +
  theme_minimal()
```

7. Selected Transformation

Logarithmic transformation (**log10**) is selected.

```
scale_x_log10()
```

8. Why Logarithmic Transformation?

The Lines of Code values range from thousands to millions. A logarithmic scale compresses very large values while spreading out smaller values.

This allows both small and large projects to be visible.

9. Advantages

- Reduces the effect of extreme values.
- Makes smaller teams easier to distinguish.
- Improves visualization of highly skewed data.
- Allows a wider range of values to be displayed.
- Helps identify relationships between Lines of Code and Bugs.

10. Observation – Nonlinear Scale

- All teams become more evenly distributed along the X-axis.
- Small projects are no longer compressed.
- The relationship between Lines of Code and Bugs becomes easier to inspect.
- Large projects remain visible without dominating the entire graph.

11. Comparison

Feature	Linear Scale	Logarithmic Scale
Small projects	Compressed	Clearly visible
Large projects	Dominant	Controlled
Skewed data	Poor representation	Better representation
Distribution	Difficult to interpret	Easier to interpret
Relationship analysis	Limited	Improved

12. Conclusion

A **logarithmic X-axis** is more appropriate for this dataset because Lines of Code are highly skewed. The nonlinear transformation improves visibility of both small and large projects and provides better analytical interpretation than a linear scale.